"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues"

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1) $\begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 3\alpha & 5 & -9\alpha & 1 \\ 2\beta & 3 & \alpha^2 & \end{bmatrix}$ $\xrightarrow{R_3 - 2R_1 \to R_3}$ $\begin{bmatrix} 1 & \alpha & 4 & \alpha \\ 3\alpha & 5 & -3\alpha & 1 \\ 0 & 2\alpha-\beta & 15 & \alpha^2-4 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

$\downarrow R_2 - 3\alpha R_4 = R_2$

$\begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 0 & 5-42\alpha & -9\alpha+18\alpha & 1-6\alpha \\ 0 & 2\alpha-\beta & 15 & \alpha^2-4 \\ 0 & 14-\alpha & -6+9 & 2-\alpha \end{bmatrix}$ $\xleftarrow{}$ $\begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 0 & 5-42\alpha & -9\alpha+18\alpha & 1-6\alpha \\ 0 & 2\alpha-\beta & 15 & \alpha^2-4 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

2)

a) minimum direction is 2 because $x = 5y$ is a plane so it has to be minimum 2D.

b) to have a volume in intersection, unknown geometry must have 3 unknows such as (x, y, z) because volume means 3D.

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3-) 1-example

$v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ in R_2

$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}$

$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$H = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

a) $H^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ so $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are unit vector.

↳ I

b) $H^T \cdot H =$ identity

$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ identity matrix.

c) $\det(A - \lambda I) = 0$

$\det \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0$ $\begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$

$\det \begin{bmatrix} -1-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix} = 0$

$(-1-\lambda) \cdot (1-\lambda) = 0$

$\boxed{\lambda_1 = 1}$ $\boxed{\lambda_2 = -1}$

d) H is diagonalizable because H is 2x2 matrix and has 2 independent eigenvectors.

$S^{-1} \Lambda S = \Lambda$

$A = S^{-1} \Lambda S$

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4) $\frac{du}{dt} = -u+w$ $\quad \frac{d}{dt}\begin{bmatrix} u \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} u \\ w \end{bmatrix}$

$\frac{dw}{dt} = u-w$

✓ no mate loss ✗ answer of 9

$$A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\det(A-\lambda I) = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{bmatrix}$$

$$(-1-\lambda)^2 - 1 = 0$$

$$\lambda_1 = -2 \quad \lambda_2 = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \cdot x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u(t) = S_1 \cdot e^{\lambda_1 t} c_1 + S_2 \cdot e^{\lambda_2 t} \cdot c_2$$

$$u(t) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot e^{-2t} c_1 + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot e^{0t} \cdot c_2$$

$t \to \infty \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot e^{-2t} \cdot c_1 + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot c_2$

$$\begin{bmatrix} u \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot c_2 \quad \begin{matrix} \%50 \ u \\ \%50 \ w \end{matrix} \rightarrow \text{answer of c}$$

b) $t=1 \quad u(1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2} \cdot c_1 + \begin{bmatrix} 1 \\ 1 \end{bmatrix} c_2$

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